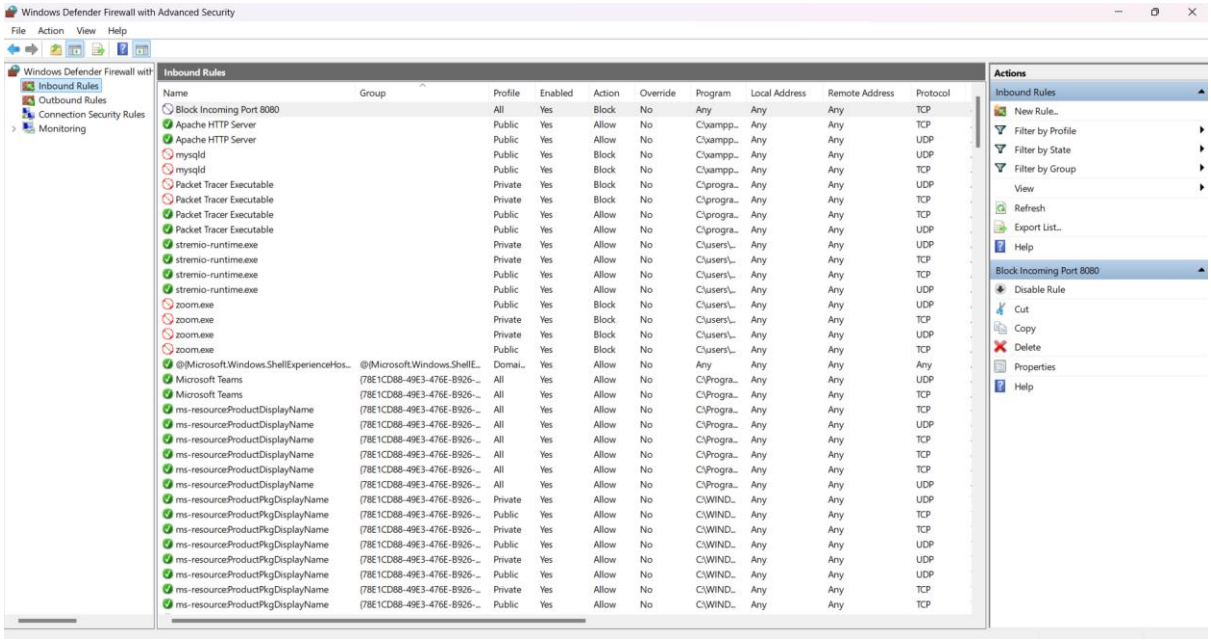
Task 1:

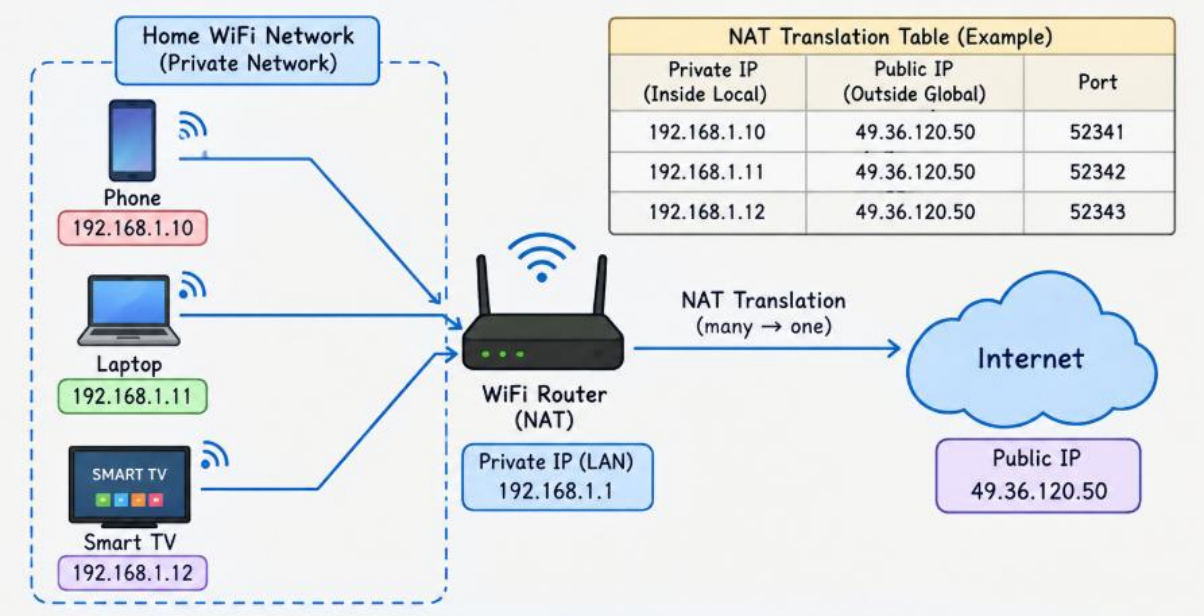
Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



Task 2:

Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WIFI router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Answer:



The home WIFI router assigns private IP addresses to devices such as a phone, laptop, and smart TV.

These devices use private IP addresses like 192.168.1.10, 192.168.1.11, and 192.168.1.12.

The router uses NAT (Network Address Translation) to translate these private addresses into one public IP address, such as 49.36.120.50, when accessing the Internet.

This allows multiple devices in the home network to share a single public IP address.

Task 3:

Configure port filtering using your WIFI router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Answer:

Steps to Configure Port Filtering

1. Connect the laptop or PC to the WIFI router.
2. Open a web browser such as Chrome or Edge.
3. Enter the router's IP address in the address bar, usually 192.168.1.1 or 192.168.0.1.
4. Log in to the router's Admin Panel using the administrator username and password.
5. Open Advanced Settings, Security, or Firewall settings.
6. Find the Port Filtering, Access Control, or Traffic Rules option.
7. Click Add/New Rule.
8. Configure the rule as follows:
    - Rule Name: Block Outgoing SMTP
    - Direction: Outgoing / LAN to WAN
    - Protocol: TCP
    - Destination Port: 25
    - Action: Block/Deny
9. Save the rule and click Apply.
10. The router will now block outgoing TCP traffic using port 25.

How This Helps Prevent Spam

Port 25 is traditionally used for SMTP (Simple Mail Transfer Protocol) to send email between mail servers. Blocking outgoing port 25 prevents devices on the home network from directly sending email through this port, which can reduce spam if a computer or other device is infected with malware. Legitimate email services can still use authenticated submission ports such as 587 or 465, depending on the email provider.

Task 4:

Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Answer:

Packet	**Source IP**	**Destination IP**	**Port**	**Result**
1. HTTP Request	192.168.1.5	172.217.166.78	80	**Allowed**
2. SSH Request	10.0.0.10	34.201.23.1	22	**Blocked**
3. DNS Query	192.168.1.7	8.8.8.8	53	**Allowed**

HTTP request: The packet uses destination port **80**, which is normally allowed for HTTP traffic, so it is allowed if the rule permits HTTP.

SSH request: The packet uses destination port **22**. If SSH traffic is not permitted by the firewall rule, the packet is blocked.

DNS query: The packet is sent to **8.8.8.8** on port **53**, which is used for DNS. If DNS traffic is permitted, the packet is allowed.

Task 5:

Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Answer:

3 Ways Instagram or Zomato Might Use Firewalls and Packet Filtering

Instagram and Zomato handle a very large amount of Internet traffic, so firewalls and packet filtering can be used as an important layer of network security. Packet-filtering rules can examine source/destination IP addresses, ports, and protocols and then allow or block traffic according to security policies.

1. Block malicious IP addresses
They can maintain blocklists of known malicious or suspicious IP addresses and prevent those systems from connecting to important servers. This can help reduce unauthorized access, scanning, and other malicious traffic.

2. Allow only required ports and protocols
Firewalls can allow required services such as **HTTPS (TCP port 443)** while blocking unnecessary or risky ports. This reduces the number of network services exposed to attackers and therefore reduces the attack surface.

3. Protect against DoS/DDoS and excessive traffic
Firewalls and network filtering can detect or block abnormal traffic such as large TCP/UDP floods, suspicious connection attempts, or traffic coming from known attack sources. DDoS protection commonly uses traffic filtering and rate limiting to keep legitimate users' traffic available during an attack.